Exp No:11 Experiment to understand pandas library use cases in Data science.

Description: Understand data frame use cases using pandas library

Code:

```
import numpy as np
import pandas as pd
list=[[1,'Smith',50000],[2,'Jones',60000]]
df=pd.DataFrame(list)
df
```

	0	1	2
0	1	Smith	50000
1	2	Jones	60000

```
df.columns=['Empd','Name','Salary']
df
```

	Empd	Name	Salary
0	1	Smith	50000
1	2	Jones	60000

```
df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2 entries, 0 to 1
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   Empd    2 non-null      int64
 1   Name    2 non-null      object
 2   Salary  2 non-null      int64
dtypes: int64(2), object(1)
memory usage: 180.0+ bytes
df = pd.read_csv('50_Startups.csv')
```

df.info()

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 50 entries, 0 to 49

Data columns (total 5 columns):

#	Column	Non-Null Count	Dtype
0	R&D Spend	50 non-null	float64
1	Administration	50 non-null	float64
2	Marketing Spend	50 non-null	float64
3	State	50 non-null	object
4	Profit	50 non-null	float64

dtypes: float64(4), object(1)

memory usage: 2.1+ KB

df.head()

	R&D Spend	Administration	Marketing Spend	State	Profit
0	165349.20	136897.80	471784.10	New York	192261.83
1	162597.70	151377.59	443898.53	California	191792.06
2	153441.51	101145.55	407934.54	Florida	191050.39
3	144372.41	118671.85	383199.62	New York	182901.99
4	142107.34	91391.77	366168.42	Florida	166187.94

df.tail()

	R&D Spend	Administration	Marketing Spend	State	Profit
45	1000.23	124153.04	1903.93	New York	64926.08
46	1315.11	110716.75	197661.10	Florida	42559.73
47	0.00	154146.00	0.00	California	35673.41
48	542.05	51743.15	0.00	New York	35673.41

	R&D Spend	Administration	Marketing Spend	State	Profit
49	0.00	116983.80	45173.06	California	14681.40

df= pd.read_csv('/content/employee.csv')

df.head()

	emp id	name	salary
0	1	SREE VARSSINI K S	5000
1	2	SREEMATHI B	6000
2	3	SREYA G	7000
3	4	SREYASKARI MULLAPUDI	5000
4	5	SRI AKASH U G	8000

df.tail()

	emp id	name	salary
5	6	SRIHARSHA V	6500
6	7	SRINITHI S	5500
7	8	SRIVATSAN R	7500
8	9	SUBHASHINI M	8500
9	10	SUDHARSHAN R	9000

df.info()

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 10 entries, 0 to 9

Data columns (total 3 columns):

 # Column Non-Null Count Dtype

--- ------ -------------- -----

 0 emp id 10 non-null   int64

 1 name   10 non-null   object

 2 salary 10 non-null   int64

dtypes: int64(2), object(1)

memory usage: 372.0+ bytes

df.salary

	salary
0	5000
1	6000
2	7000
3	5000
4	8000
5	6500
6	5500
7	7500
8	8500
9	9000

dtype: int64

type(df.salary)

df.salary.mean()

np.float64(6800.0)

df.salary.median()

6750.0

df.salary.var()

2066666.6666666667

df.salary.std()

1437.5905768565217

df.describe()

	emp id	salary
count	10.00000	10.000000
mean	5.50000	6800.000000

	emp id	salary
std	3.02765	1437.590577
min	1.00000	5000.000000
25%	3.25000	5625.000000
50%	5.50000	6750.000000
75%	7.75000	7875.000000
max	10.00000	9000.000000

df.describe(include='all')

	emp id	name	salary
count	10.00000	10	10.000000
unique	NaN	10	NaN
top	NaN	SREE VARSSINI K S	NaN
freq	NaN	1	NaN
mean	5.50000	NaN	6800.000000
std	3.02765	NaN	1437.590577
min	1.00000	NaN	5000.000000
25%	3.25000	NaN	5625.000000
50%	5.50000	NaN	6750.000000
75%	7.75000	NaN	7875.000000
max	10.00000	NaN	9000.000000

```
empCol=df.columns

empCol

Index(['emp id', 'name', 'salary'], dtype='object')

emparray=df.values

emparray

array([[1, 'SREE VARSSINI K S', 5000],
       [2, 'SREEMATHI B', 6000],
```

[3, 'SREYA G', 7000],

[4, 'SREYASKARI MULLAPUDI', 5000],

[5, 'SRI AKASH U G', 8000],

[6, 'SRIHARSHA V', 6500],

[7, 'SRINITHI S', 5500],

[8, 'SRIVATSAN R', 7500],

[9, 'SUBHASHINI M', 8500],

[10, 'SUDHARSHAN R', 9000]], dtype=object)

```
employee_DF=pd.DataFrame(emparray,columns=empCol)
```

employee_DF

	emp id	name	salary
0	1	SREE VARSSINI K S	5000
1	2	SREEMATHI B	6000
2	3	SREYA G	7000
3	4	SREYASKARI MULLAPUDI	5000
4	5	SRI AKASH U G	8000
5	6	SRIHARSHA V	6500
6	7	SRINITHI S	5500
7	8	SRIVATSAN R	7500
8	9	SUBHASHINI M	8500
9	10	SUDHARSHAN R	9000

Result:

Understand data frame use cases using pandas library